Homework 5

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Section 6.1

1) b) $B \subseteq A$ and $B \subset A$
   d) $B \subseteq A$ and $B \subset A$

4) $B \subseteq A$ because every integer that is divisble by 20 is also divisible by 5:
   - $m = 20s,\ s \in \mathbb{Z}$,
   - $m = 5(4s),\ s \in \mathbb{Z}$
   - $m = 5r,\ r \in \mathbb{Z}$ because integers are closed under addition

   $A \nsubseteq B$, Example : $5 \in A, 5 \notin B$

12) a) $\{x \in \mathbb{R} \mid -3 \leq x < 2\}$
    b) $\{x \in \mathbb{R} \mid -1 < x \leq 0\}$
    g) $\{x \in \mathbb{R} \mid x < -3 \text{ or } x \geq 2\}$
    i) $\{x \in \mathbb{R} \mid x \leq -1 \text{ or } x > 0\}$

25) a) $[1, 2]$
    c) No, $R_1, R_2, ...$ are not mutually disjoint because they all contain the value 1 and a small range greater than 1.
    d) $[1, 2]$
    f) $[1, 2]$

29) <u>Yes</u>, $\{\mathbb{R}^+, \mathbb{R}^-, \{0\}\}$ is a partition of $\mathbb{R}$ because
    - $\mathbb{R}^+ \cup \mathbb{R}^- \cup \{0\} = \mathbb{R}$
    - $\mathbb{R}^+, \mathbb{R}^-,$ and $\{0\}$ are mutually disjoint because they contain none of the same values.

Section 9.2

11) c) There are 6 positions that can be manipulated, each with 2 possible values. Therefore, by the multiplication rule:

$$\boxed{\therefore \text{ There are } 2^6 = 32 \text{ possible bit strings of length 8 that begin and end with a 1}}$$

17) c)

Digit #	# Possible Digits
1	9
2	9
3	8
4	7

$$\boxed{\therefore \text{ By the multiplication rule, there are } 9 \cdot 9 \cdot 8 \cdot 7 = 4536 \text{ possible integers from 1000 through 9999 with distinct digits.}}$$

e)
- $P_r(\text{Distinct Digits}) = \frac{\text{\# Distinct Digits}}{\text{Total \#}} = \frac{4536}{9000} = \frac{63}{125}$

Digit #	# Possible Digits
4	5
1	8
2	8
3	7

- $P_r(\text{Distinct Digits and Odd}) = \frac{\text{\# Distinct Digits and Odd}}{\text{Total \#}} = \frac{2240}{9000} = \frac{56}{225}$

21) a) The number of relations from A to B is the number of possible sets of ordered pairs $(x, y),\ x \in A, y \in B$. There are $m \cdot n$ possible pairs, by the multiplication rule. Every ordered pair, (x, y) can either be present in the set, or absent. Since there $m \cdot n$ pairs and every pair has two possible states, by the multiplication rule,

$$\boxed{\therefore \text{ There are } 2^{mn} \text{ possible relations from } A \text{ to } B}$$

b) The number of functions from A to B is the number of possible mappings from every element in A to every element in B.

Element $a \in A$	# Possible Elements $b \in B$
a_1	n
a_2	n
...	...
a_m	n

Therefore, by the multiplication rule,

$$\boxed{\therefore \text{ There are } n^m \text{ possible functions from } A \text{ to } B}$$

c) $\frac{m^n}{2^{mn}}$

31) d)

Distinct Prime p_i	# Possible Divisors
p_1	$a_1 + 1$
p_2	$a_2 + 1$
...	...
p_m	$a_m + 1$

Therefore, by the multiplication rule,

$$\boxed{\therefore \text{ There are } \prod_{i=1}^{m}(a_i + 1) \text{ distinct positive divisors in } \prod_{i=1}^{m} p_i^{a_i}}$$

e) Possible ways to get 12 divisors: $(a_1 + 1)(a_2 + 1)... = 12$, where p are the smallest distinct prime numbers.
- $p^{11} : 2^{11} = 2048$
- $p^5 p^1 : 2^5 \cdot 3 = 96$
- $p^3 p^2 : 2^3 \cdot 3^2 = 72$
- $p^2 p^1 p^1 : 2^2 \cdot 3 \cdot 5 = 60$

$$\boxed{\therefore \text{ The smallest positive integer with 12 positive integer divisors is 60}}$$

39) b) $P(9, 6) = \frac{9!}{3!} = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 60480$
d) $P(7, 4) = \frac{7!}{3!} = 7 \cdot 6 \cdot 5 \cdot 4 = 840$

Section 9.3

2) b)

String Length	# Possible Hexadecimal Numbers
2	$16 \cdot 16$
3	16^3
4	16^4
5	16^5

By the addition rule,

$\therefore$ There are $16^2 + 16^3 + 16^4 + 16^5 = 1118464$ possible strings of hexadecimal digits that consist of two to five digits.

7) c)

Password Length	# Possible Passwords
3	50^3
4	50^4
5	50^5

The number of total possible passwords, by the addition and multiplication rules is $50^3 + 50^4 + 50^5 = \underline{318875000}$

Password Length	# Possible Passwords; No Repetitions
3	$50 \cdot 49 \cdot 48$
4	$50 \cdot 49 \cdot 48 \cdot 47$
5	$50 \cdot 49 \cdot 48 \cdot 47 \cdot 46$

The number of possible passwords with no repetitions is $50(49)(48) + 50(49)(48)(47) + 50(49)(48)(47)(46) = \underline{259896000}$

$\therefore$ Therefore by the difference rule, the number of possible passwords with at least 1 repetition is $318875000 - 259896000 = 58979000$

d) $P_r(\text{At Least 1 Repetition}) = \frac{58979000}{318875000} \approx 0.185$

17) a) $16 \cdot 15 \cdot 14 \cdot 13 = 43680$ four digit hexadecimal strings with no repetitions.

b) # 4-digit strings at least 1 repetition = # total strings − # strings no repetitions $= 16^4 - 43680 = \underline{21856}$

c) $P_r(\text{At least 1 repetition}) = \frac{21856}{16^4} = \frac{683}{2048} \approx 0.333$

23) a) $\left\lfloor \frac{1000}{4} \right\rfloor + \left\lfloor \frac{1000}{7} \right\rfloor - \left\lfloor \frac{1000}{28} \right\rfloor = 357$ integers 1 through 1000 divisible by 4 or 7.

b) $P_r(\text{Divisible by 4 or 7}) = \frac{357}{1000} = 0.357$

c) # integers 1 through 1000 not divisible by 4 or 7 $= 1000 - 357 = 643$

Section 9.4

4) Yes, there are $26^2 = 676$ combinations of first and last initials. Therefore by the pigeonhole principle a function from people in the group to initials combinations is not one-to-one.

$$\boxed{\therefore \text{ There is a at least set of 2 people with the same first and last initials}}$$

7) Let there be a function $f : A \to B$ where A is the six randomly chosen integers and B be the following partition of $S : \{\{3, 12\}, \{4, 11\}, \{5, 10\}, \{6, 9\}, \{7, 8\}\}$.

Let f be a function that maps the integer in A to the corresponding subset that contains the integer in B. Because $|A| = 6 > |B| = 5$, by the pigeonhole principle, there are at least two integers that correspond to the same subset in B. Because all of the subsets in B have elements that sum to 15 and there are no repeated elements,

$$\boxed{\therefore \text{ If six random integers are chosen from } S\text{, there will be at least two integers that sum to 15.}}$$

11) Let a function $f : A \to B$ map randomly chosen integers to sets of which they are contained. A is the set of the $n + 1$ randomly chosen integers, and B is the following partition of the set: $\{\{1, 2\}, \{3, 4\}, ..., \{2n - 1, 2n\}\}$

Because $|A| = n + 1 > |B| = n$, by the pigeonhole principle, there are at least two randomly chosen integers that correspond to the same subset in B. Because there are no repeated integers, and every subset contains 1/2 even integers,

$$\boxed{\therefore \text{ At least one of the randomly chosen } n + 1 \text{ integers will be even.}}$$

31) Suppose that there are 2 assistants that are assigned to 3 or more executives– let's assume the scenario with the maximum number of executives: 2 assistants with 4 executives each. This means that the other 3 assistants are assigned to a maximum of 2 executives each. The maximum possible number of executives with assistants in this scenario is $3 \cdot 2 + 2 \cdot 4 = 14$. This contradicts the fact that there are 15 executives, therefore our assumption is false, and

$$\boxed{\therefore \text{ At least 3 assistants are assigned to at least 3 executives}}$$

33) Because there are 6 elements in A, there are $2^6 = 64$ distinct subsets of A, including $\emptyset$, and there are 63 distinct subsets that have an existing sum. Let k be the smallest element in A. This means that the minimum sum of a subset of A is simply k, and the maximum sum is $k + 10 + 11 + 12 + 13 + 14 = k + 60$. There are 61 elements in the range from k to $k + 60$ and there are 63 possible subset sums from A. Therefore by the pigeonhole principle, there are at least two subsets of A that have the same sum.

Section 9.5

7) a) $C(13,7) = \frac{P(13,7)}{7!} = \frac{13!}{7!(13-7)!} = \underline{1716}$

b) i) $C(7,4) \cdot C(6,3) = \underline{700}$

ii) $1716 - 1 = \underline{1715}$

iii) $C(7,1) \cdot C(6,6) + (7,2) \cdot C(6,5) + (7,3) \cdot C(6,4) = \underline{658}$

c) $C(13,7) - C(11,5) = \underline{1254}$

- You must subtract out all of the combinations where both members are already in the group.

d) $C(11,5) + C(11,7) = \underline{792}$

8) a) $C(14,10) = \frac{14!}{10!4!} = \underline{1001}$

b) i) $C(6,4) \cdot C(8,6) = \underline{420}$

ii) $C(14,10) - C(6,10) = \underline{1001}$

iii) $C(8,8) \cdot C(6,2) + C(8,7) \cdot C(6,3) = \underline{175}$

c) $C(12,10) + C(12,9) \cdot C(2,1) = \underline{506}$

d) $C(12,8) + C(12,10) = \underline{561}$

10) $C(60,22) \cdot C(60-22,22) \cdot C(60-22-22,16) = C(60,22) \cdot C(38,22)$
$= \frac{60!}{22!38!} \cdot \frac{38!}{22!16!} = \frac{60!}{22!22!16!} \approx 9.15 \cdot 10^{26}$

13) b) $C(10,5) = \underline{252}$

e) $C(10,0) + C(10,1) = \underline{11}$

16) b) $C(40,5) - C(37,5) = \underline{222111}$

c) $P_{r(\text{Defective})} = \frac{222111}{C(40,5)} = \frac{222111}{658008} \approx 0.338$

20) a) $\frac{11!}{3!2!2!1!1!1!1!} = \underline{1663200}$

b) $\frac{9!}{3!2!1!1!1!1!} = \underline{30240}$

c) $\frac{9!}{3!2!2!1!1!} = \underline{15120}$

Section 9.8

3) a) 0.6

b) <u>No</u>, $P(C) = 0.4$ because A, B, and C are mutually exclusive, so $P(A) + P(B) + P(C) = 1$. Thus $0.4 + 0.2 + P(C) = 1$

15)
- Total Revenue $= 3000(\$20) = \$60000$
- Total Prize Expenses $= \$40000 + \$1000 + \$500 = \$41500$
- Profit = Total Revenue − Total Prize Expenses

  $= \$60000 - \$41500 = \$18500$
- Expected Profit / Ticket $= \frac{\$18500}{3000 \text{ Tickets}} =$
- $\boxed{\approx \$6.17 \text{ to charity for each ticket sold.}}$

18)

Combination	# Ways to Select	Sum
{1,2,2}	1	5
{1,2,8}	4	11
{2,2,8}	2	12
{1,8,8}	1	17
{2,8,8}	2	18

$$\text{Expected Sum} = \sum_{i=0}^{n} a_i P_i$$
$$= 5\left(\tfrac{1}{10}\right) + 11\left(\tfrac{4}{10}\right) + 12\left(\tfrac{2}{10}\right) + 17\left(\tfrac{1}{10}\right) + 18\left(\tfrac{2}{10}\right)$$
$$\boxed{= \tfrac{63}{5} = 12.6}$$

Section 9.9

2) $P(X \mid Y) = \frac{P(X \cap Y)}{P(Y)}$

$\Rightarrow \frac{1}{3} = \frac{P(X \cap Y)}{\frac{1}{4}}$

$\boxed{P(X \cap Y) = \frac{1}{12}}$

18) Using Bayes' Formula:

$P(B \mid A)P(A) = P(A \mid B)P(B) = P(A \cap B)$

$\Rightarrow P(B \mid A)P(A) = P(A \mid B)P(B) = P(B) \cdot P(A)$

$\Rightarrow \frac{P(B \mid A)P(A)}{P(A)} = P(B), \ \frac{P(A \mid B)P(B)}{P(B)} = P(B)$

$\Rightarrow \boxed{\therefore P(B \mid A) = P(B), P(A \mid B) = P(A)}$

Additional Questions

Blue and Gray Dice:

Rolls with Sum of 6: **5**

- $\{1, 5\}$
- $\{2, 4\}$
- $\{3, 3\}$
- $\{4, 2\}$
- $\{5, 1\}$

$P(\text{Sum} = 6) = \frac{5}{36}$

Rolls with Both Odds: **9**

- $\{1, 1\}$
- $\{1, 3\}$
- $\{1, 5\}$
- $\{3, 1\}$
- $\{3, 3\}$
- $\{3, 5\}$
- $\{5, 1\}$
- $\{5, 3\}$
- $\{5, 5\}$

$P(\text{Both Odd}) = \frac{9}{36} = \frac{1}{4}$

Rolls with Sum of 6 and Both Odds: **3**

- $\{1, 5\}$
- $\{3, 3\}$
- $\{5, 1\}$

$$P(\text{Sum} = 6 \cap \text{Both Odd}) = \frac{3}{36} = \frac{1}{12}$$

$$\Rightarrow P(\text{Sum} = 6 \mid \text{Both Odd}) = \frac{P(\text{Sum} = 6 \cap \text{Both Odd})}{P(\text{Both Odd})}$$

$$\boxed{P(\text{Sum} = 6 \mid \text{Both Odd})\frac{\frac{1}{12}}{\frac{1}{4}} = \frac{1}{3}}$$

98% Accurate Drug Test:

We are trying to find $P(\text{Drug User} \mid \text{Positive Test})$

- Bayes' Formula: $P(B \mid A)P(A) = P(A \mid B)P(B) = P(A \cap B)$

$$\Rightarrow P(\text{Drug User} \mid \text{Positive Test}) \cdot P(\text{Positive Test}) = P(\text{Positive Test} \mid \text{Drug User}) \cdot P(\text{Drug User})$$

$$\Rightarrow P(\text{Drug User} \mid \text{Positive Test}) = \frac{P(\text{Positive Test} \mid \text{Drug User}) \cdot P(\text{Drug User})}{P(\text{Positive Test})}$$

- $P(\text{Positive Test} \mid \text{Drug User}) = 98\% = 0.98$
- $P(\text{Drug User}) = 0.5\% = 0.005$
- $P(\text{Positive Test}) = 0.5\% \cdot 98\% + 99.5\% \cdot 2\% = 2.48\% = 0.0248$

$$\boxed{\Rightarrow P(\text{Drug User} \mid \text{Positive Test}) = \frac{0.98 \cdot 0.005}{0.0248} = \frac{49}{248} \approx 0.198}$$